Telescoping product (7.14)

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(Telescoping product.) Prove that

$$\frac{1}{15} < \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\frac{7}{8}\cdots\frac{99}{100} < \frac{1}{10}.$$

Hint :

$$(1)\ A = \frac{1}{2}\cdot\frac{3}{4}\cdots\frac{99}{100}, \qquad (2)\ A < \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{100}{101}, \qquad (3)\ A > \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}\cdots\frac{98}{99}.$$

Multiply (1) with (2) and (1) with (3).